PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: InvenIQ

Version: 1.0

1. Project Overview

- **Problem Statement:** Small and Medium Enterprises (SMEs) in Malaysia often operate on thin margins and rely heavily on intuition-based decision-making. Fragmented operational data from sales, inventory, and market signals are rarely consolidated into actionable intelligence. This results in:
    - Overstocking, causing capital lock-up
    - Understocking, leading to lost revenue opportunities
    - Inefficient pricing and procurement strategies
    - Professional financial analytics solutions remain costly and inaccessible for micro-SMEs, limiting their competitiveness against larger data-driven corporations.
- **Target Domain:** AI-driven Decision Intelligence for Economic Empowerment, specifically focusing on SME Operational Efficiency and Financial Forecasting.
- **Proposed Solution Summary:** InvenIQ is an AI-powered conversational inventory strategist designed for SMEs. The system integrates:
    - Structured business data (CSV sales logs)
    - Unstructured contextual inputs via a **Context Sidebar** including:
        - Weather trends
        - Local calendar events
        - Market news
        - Raw business communications (e.g., WhatsApp supplier messages)

Instead of relying on proprietary hosted AI infrastructure, the system now utilizes **high-performance inference through the Groq API**, powered by the **Groq platform** and the **Llama 3.3 70B Versatile model**, enabling fast, cost-efficient reasoning and conversational analytics.

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2. Background & Business Objective

- **Background of the Problem:** Inventory management is often rigid and requires special skills or expensive software. SMEs owners struggle to optimize their inventory to increase revenue because they lack these specialized tools. This manual process requires extensive labor work, specialized software and the skills, which are limited to owners from seeing new ways to improve their business.

- **Importance of Solving this Issue:** By reducing analysis time from days to seconds, SMEs can rapidly test operational strategies and evaluate business scenarios. Using Groq-hosted large language models removes technical complexity, allowing users to interact with business intelligence systems through natural language instead of technical dashboards. The user's role evolves from **manual operator** to **strategic decision curator**.

- **Strategic Fit / Impact:** This project aligns with the AI-Native development lifecycle by providing an AI-enabled partner for business decisions. The system acts as a high-speed generative engine that understands store needs through natural language prompts. It helps entrepreneurs and founders focus on growth by making complex data easy to utilize.

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3. Product Purpose

3.1. Main Goal of the System: InvenIQ is designed to be an AI-powered Decision Intelligence partner that transforms operational data into actionable economic strategies.

Powered by Groq's ultra-low-latency inference, the platform delivers explainable recommendations, inventory optimization insights, and profit-oriented decision support.

Unlike systems that only prescribe actions, InvenIQ emphasizes explaining why a recommendation is made, ensuring transparency and trust in every decision.

3.2. Intended Users (Target Audience):

- SME Owners: Individuals running small retail or service businesses who lack formal financial training.
- Retail Managers: Decision-makers responsible for inventory procurement and stock-level management.
- Financial Analysts: Users who need to generate rapid insights from multi-source data (structured sales logs + unstructured market trends) for their clients.
- Operations Leads: Individuals focused on reducing overhead costs and improving supply chain efficiency.
- Gig Economy Hub Managers: Community-level coordinators managing resource allocation and task scheduling for local worker groups.

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4. System Functionalities

4.1. Description: The system operates as a conversational analytics engine. Users upload structured data and provide natural language queries to receive optimized business plans.

4.2. Key Functionalities:

- Automated Forecasting: Predicts demand based on historical sales and seasonal trends.
- Strategic Recommender: Suggests specific reorder points and discount levels to clear slow-moving stock.
- Natural Language Insights: Explains the "why" behind data trends (e.g., "Sales dropped due to unseasonably rainy weather"

4.3. AI Model & Prompt Design

- **4.3.1. Model Selection:** Groq API is the core engine, chosen for its superior ability to reason across structured datasets and generate human-readable strategic explanations.
- **4.3.2. Prompting Strategy:** Multi-step agentic prompting is used to first analyze the data, then cross-reference it with user intent, and finally generate the recommendation.
- **4.3.3. Context & Input Handling:** The system uses chunking for oversized CSV files, processing data in segments to maintain context within the model's token limit. Maximum input size is set at 2,000 rows per analysis.
- **4.3.4. Fallback & Failure Behavior:** If the model returns a hallucinated or off-topic response, the system triggers a graceful error state and asks the user to refine the prompt or re-upload the data.

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5. User Stories & Use Cases

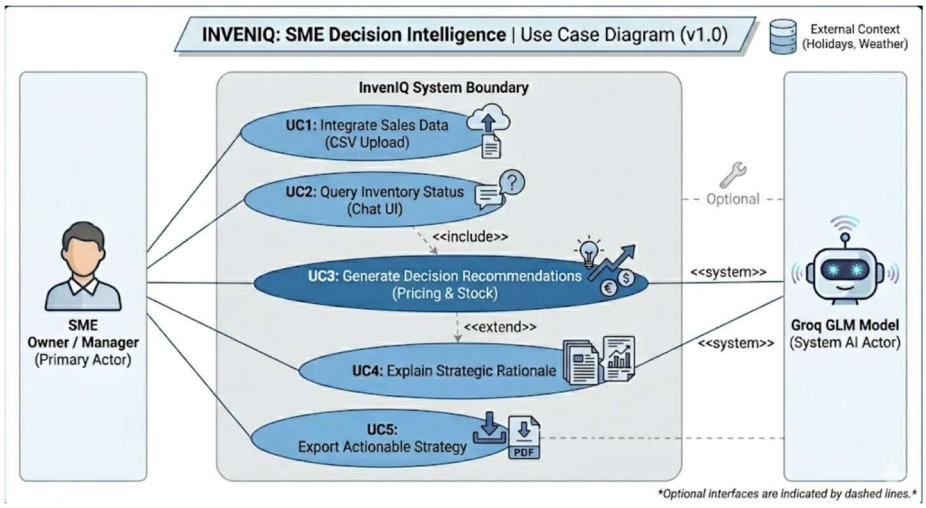
- **User Stories:** As an SME Owner, I want to enter a business intent in my own words (e.g., "Clear out old stock to make room for new arrivals") so that it provides me with an optimized inventory layout without me having to manually calculate stock levels.

- **Use Cases (Main Interactions):**

  Prompt to UI Flow: Users enter a strategic prompt (e.g., *"Reduce storage costs for slow-moving items"*). The system then interprets the business intent using the Groq API powered by the Llama 3.3 70B Versatile model, identifies relevant inventory components and turnover variables, and renders a customizable 2D Strategy Canvas displaying the optimized stock layout.

  Flow Generation: User selects a specific inventory component or bottleneck on the canvas and triggers/enables the section called "Generate Next Scenario". generates the target outcome (e.g., a flash sale flow or a reorder sequence) while maintaining the business's digital tokens, such as profit margin targets, budget constraints, and brand preferences.

- **Use Cases Diagram**



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6. Features Included (Scope Definition)

- **Data Upload Portal:** Supports CSV/Excel files for structured sales data.
- **Conversational Dashboard:** A chat interface for querying the data.
- **Strategy Export:** One-click export of recommended reorder lists to PDF

7. Features Not Included (Scope Control)

- **Direct ERP Integration:** Initial version will not connect directly to 3rd party inventory systems.
- **Real-time IoT Tracking:** Does not support real-time sensor-based inventory monitoring.

8. Assumptions & Constraints

- **LLM Cost Constraint:** Maximum token cost is 1024 tokens per response. To manage this, we implement caching of repeated analytical queries.
- **Technical Constraints:** Analysis is limited to structured data formats (CSV/XLS) and text-based market news.
- **User Input:** The system requires a minimum of 3 months of historical data to generate accurate forecasts.

9. Risks & Questions Throughout Development

- **Data Accuracy:** How do we ensure the AI doesn't hallucinate numbers when summarizing large datasets?
- **Economic Impact:** How will we quantify the "revenue improvement" for a simulated scenario during the preliminary round?

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